Machine Learning Based Severity Classification of Sickle Cell Disease Patients using Clinical Data from the MIMIC-IV Critical Care Dataset

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Abstract: -

Sickle Cell Disease (SCD) is an inherited blood disorder that alters the form and function of red blood cells, causing potential severe health issues. Having the ability to forecast the potential severity of the disease can help doctors take earlier and more effective measures. For this research, we employed the MIMIC-IV critical care dataset to train and compare three machine learning models: XGBoost, Random Forest, and Logistic Regression to predict the severity of SCD. Before training, we ensured that the dataset was thoroughly cleaned and preprocessed to address missing values, categorical variables, scaling numerical variables, and addressing class imbalance issues using RandomOverSampler. Among all models, XGBoost performed best with a test accuracy of 95.6% and produced accurate results for all levels of severity, indicating that the model generalizes well and does not overfit. These results indicate that machine learning models, when Fine-tuned and balanced, can enable doctors to make quicker. This approach has the potential to assist healthcare professionals in prioritizing high-risk patients and optimizing treatment strategies. Future work may focus on incorporating additional clinical variables, validating the models on external datasets, and developing an interactive clinical decision-support dashboard for real-world use.

1. Introduction: -

Sickle Cell Disease (SCD) is a genetic disorder marked by changes in hemoglobin. This changes creates abnormal, rigid red blood cells, which blocks oxygen delivery, and characterized by chronic haemolytic anaemia and episodic Vaso-occlusive crises, which leads to significant illness and death worldwide [11]. The severity of Sickle Cell Disease (SCD) often varies unpredictably, which makes timely treatment and effective management difficult, particularly in healthcare systems with limited resources [12]. With the growing adoption of electronic health records, machine learning techniques offer an opportunity to support doctors by identifying patterns that may not be immediately visible through traditional methods. The prediction of SCD severity, might help clinicians in decision- making and reducing preventable complications [13]. Predictive models can also contribute to national public health initiatives, such as the National Sickle Cell Anaemia Elimination Mission (NSCAEM), by helping identify high-risk patients and prioritizing care interventions [14] Despite these opportunities, there is a only a few number of studies applying ML models to large, real-world ICU datasets, such as MIMIC-IV, for SCD severity prediction [1],[2]. Using, the MIMIC-IV dataset presents several difficulties: some laboratory features contain missing values, not all clinically relevant variables are consistently available. Class imbalance exists across severity levels. Addressing these limitations required various preprocessing techniques , including feature selection, encoding, scaling, and handling class imbalance, to ensure robust model performance.

This study addresses this gap by:

- Building machine learning models to predict SCD severity using the MIMIC-IV critical care dataset.
- Performing a comparative evaluation of three models: XGBoost, Random Forest, and Logistic Regression to identify the most robust and generalizable classifier.

2. RELATED WORK

Research in recent years has increasingly applied machine learning techniques to hematological disorders, including Sickle Cell Disease (SCD). Early approaches relied on clinical risk scores derived from laboratory parameters and patient histories to predict complications or hospitalization [3]. Although these models were easy to interpret, they lacked generalizability and failed to capture complex patient variability.

Subsequent research adopted supervised ML algorithms such as Decision Trees and Support Vector Machines (SVMs) to predict outcomes like vaso-occlusive crises and hospitalization risk, achieving moderate success. More recently, ensemble-based approaches, particularly Random Forest and Gradient Boosting techniques, have demonstrated improved stability and predictive strength by aggregating multiple weak learners. [5].

Recent efforts have also explored deep learning and time-series models to process longitudinal patient data and unstructured clinical text [4], achieving higher accuracy but at the cost of interpretability.

This current study extends existing literature by focusing on severity classification in SCD using a hybrid model combining Random Forest and Logistic Regression. The novelty lies in its integration of predictive modelling within a web-based clinical tool, offering interpretability and real-world applicability for healthcare practitioners.

3. LITERATURE SURVEY

Sickle Cell Disease (SCD) is a hereditary blood disorder caused by abnormal hemoglobin (HbS), leading to vaso-

occlusive events, organ damage, and increased mortality [11],[12]. Traditional clinical scoring systems often fail to model the nonlinear interactions between demographics, lab biomarkers, and disease complications [10].

Recent advances have demonstrated that machine learning can substantially improve prediction of disease severity, hospitalization, and mortality in SCD patients [3]–[5]. The MIMIC-IV critical care database, although not SCD-specific, provides a rich, publicly available dataset of patient demographics, lab results, and ICU outcomes [1],[2]. Studies leveraging MIMIC-IV have shown strong predictive performance in ICU mortality and other critical conditions [6],[7], implying its potential applicability in SCD risk modelling.
Despite this promise, significant challenges persist — including limited disease-specific data, imbalanced datasets, and the need for interpretability in clinical ML models. Recent work on explainable AI (XAI), has improved model transparency and clinician trust [8]. Future research is expected to integrate multi-source datasets, including EHRs and genomic data, and adopt hybrid ensemble–deep learning frameworks to create clinically deployable, explainable decision support systems for SCD management.

Table 1: Summary table

Comparison of ML-based approaches for SCD and ICU outcome prediction

References	Dataset Used	Models / Algorithms	Best Accuracy	Key Contributions	Limitations
Mohammed et al., 2020	ICU dataset of SCD patients	Random Forest, SVM	95.06%	Developed ML model to predict acute organ failure early in ICU SCD patients	Limited dataset size; lacked interpretability
Elsabagh et al., 2024	Longitudinal Blood Pathology Data	Deep Neural Network	98.5%	Used temporal data to model disease progression and severity	Data imbalance; model complexity
Patel et al., 2021	SCD hospital records	Logistic Regression, RF, XGBoost	96.8%	Predicted hospital readmissions with ensemble ML models	Focused on single-center data
Yu et al., 2025	MIMIC-IV (general ICU dataset)	Explainable Gradient Boosting	94.2%	Applied SHAP-based explainable ML for 28-day ICU mortality	Not SCD-specific
Mesinovic et al., 2025	MIMIC-IV + eICU	Ensemble ML	93.7%	Combined multi-center ICU datasets for mortality prediction	High computational cost
Rajkomar et al., 2018	EHR (multi-hospital)	Deep Learning (RNN, CNN)	97.6%	Demonstrated scalable and accurate deep learning on EHRs	Lack of transparency in deep models
Serjeant (1997), Quinn et al. (2010)	Clinical SCD cohorts	Statistical methods	--	Identified hematological indicators of disease prognosis	Non-ML approach; limited scope
Shickel et al., 2018	EHR datasets	Review (Deep EHR frameworks)	--	Surveyed recent DL advances in EHR analytics	Did not focus on SCD specifically

4. PROPOSED METHODOLOGY: -

A. Dataset

A subset of the MIMIC-IV critical care database was used in this study, which is a large, publicly available, and de-identified dataset containing comprehensive electronic health records of ICU patients [1], [2]. Patient records were filtered using the ICD-10 code D57, corresponding to Sickle Cell Disease (SCD).

The resulting dataset included key demographic and laboratory parameters such as age, gender, hemoglobin (g/dL), hematocrit (%), and platelet count, as these are clinically significant indicators for evaluating disease progression and severity in SCD patients [11], [12].

The target variable, "Severity," was derived based on hemoglobin concentration, following WHO guidelines [9]:

- Mild: Hemoglobin > 10 g/dL
- Moderate: Hemoglobin between 7–10 g/dL
- Severe: Hemoglobin < 7 g/dL

This classification allowed the model to predict SCD severity levels for individual patients.

B. Data Imbalance Handling

The dataset exhibited class imbalance, with the "Severe" category representing a smaller proportion of samples compared to "Mild" and "Moderate." This imbalance can bias model training and lead to poor generalization.

To address this issue, Random Oversampling was

applied to increase the number of minority class samples until all classes were balanced. This approach was preferred over under sampling, as retaining all majority clas samples preserves valuable clinical information while improving the model.

After resampling, the dataset achieved a nearly uniform class distribution. As shown below

Severity Level	**Samples (Before Balancing)**	**Samples (After Balancing)**
Mild	527	527
Moderate	46	527
Sever	22	527

C. Data Preprocessing

To ensure consistency and reliable model performance, a structured preprocessing pipeline was applied applied:

1. Handling Missing Values: Missing entries were replaced using the median value for numerical variables to minimize bias.
2. Encoding Categorical Variables: Binary encoding was applied to gender (0 = female, 1 male) to convert categorical data into numerical format suitable for model training
3. Feature Scaling: Numerical variables were normalized using StandardScaler, ensuring that variables contribute equally to model learning.
4. Train-Test Split: Processed data set was then divided according to 80-20 rule to evaluate the generalization.

D. Model Development

Three supervised learning models were implemented int this study: Logistic Regression, Random Forest, and Extreme Gradient Boosting(XGBoost). These algorithms were chosen to provide a comparative perspective, starting from a simple linear baseline to more advance approaches.

Logistic Regression is a linear classification model that estimates the probability that a given input $x$belongs to a particular class. As a linear model, it estimates the probability of a sample belonging to a specific clss based on a weighted combination of input features. Sigmoid function is used to transform linear output into a value between 0 and 1.

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n)}}$$

where $\beta_0$is the intercept, $\beta_i$are the feature coefficients, and $x_i$represent the input features. In this study, Logistic Regression served as a baseline model due to its interpretability and efficiency in handling binary or multiclass classification problems.

B. Random Forest

Random Forest is an ensemble-based learning method that constructs multiple decision trees during training and aggregates their outputs to produce a final prediction. Each tree is trained on a randomly sampled subset of the data and features, introducing diversity that reduces variance and overfitting. The final output is determined through majority voting across individual tree predictions.

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_k(x)\}$$

where $h_i(x)$ is the prediction of I th tree.

This model helps capture nonlinear feature interactions and provides better generalization on imbalanced datasets after resampling.

C. Extreme Gradient Boosting (XGBoost)

XGBoost is a powerful gradient-boosted ensemble algorithm that builds trees sequentially, with each new tree minimizing the residual errors of the previous ones. It optimizes the following objective function:

$$Obj = \sum_{\{i=1\}}^{\{n\}} l(y_i, \hat{}\{y\}_i) + \sum_{\{k=1\}}^{\{K\}} \Omega(f_k)$$

where l is loss function, and
$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \mid\mid w \mid\mid^2$ is a regularization term to

prevent overfitting, with T is number of leaves.
XGBoost was employed due to its efficiency, built-in regularization, and robustness in handling complex data patterns.

- Moderate: Patients with hemoglobin levels between 7 and 10 g/dL have clinically severe

  anemia and a higher risk of problems.
- Severe: Hb < 7 g/dL: Patients are at a high risk
- of organ failure, repeated crises and ICU.

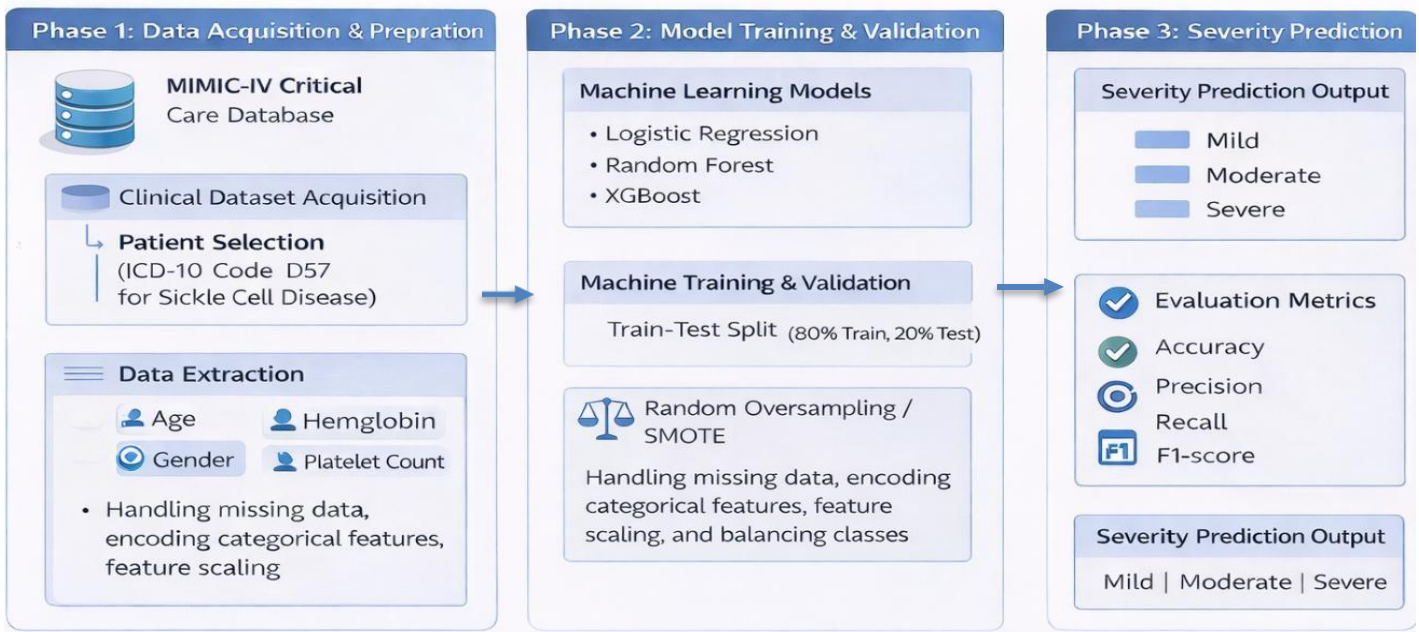

Figure 1: Overview of the SCD severity prediction process using machine learning models.

D. Justifying the Severity Definition:

Hemoglobin (Hb) concentration levels were used in this study to classify the severity of sickle cell disease (SCD), as hemoglobin is a clinically recognized biomarker to measure anemia and disease burden in SCD patients [9], [10]. In order to verify consistency with clinical practice, severity categories were modified using NIH SCD care protocols [10] and WHO anemia recommendations [9].

The severity levels were determined as follows:

- Mild: Hb > 10 g/dL: Patients usually stay stable, have a decreased risk of hospitalization, and experience fewer vaso-occlusive crises.

5. Result: -

Model performance was evaluated using accuracy, macro recall, and F1-score, assessed on both cross-validation folds and the held-out test set to ensure robust generalization. Following the application of RandomOverSampler to address class imbalance, noticeable improvements were observed, particularly in the classification of minority severity categories.

Table 1 presents a comparative summary of the final evaluation results for XGBoost, Random Forest, and Logistic Regression.

A. Model Performance

Table . Performance Metrics of ML Models for SCD Severity Prediction

Model	Accuracy	Macro Recall	F1 Score
XGboost	0.96	0.92	0.96
RandomForest	0.9565	0.92	0.96
Logistic Regression	0.94	0.91	0.94

Model Comparison Table

Insights and Observations:

The introduction of class balancing techniques notably enhanced the performance of all models, particularly for the Moderate and Severe severity classes that were previously underrepresented. Ensemble models, demonstrated High performance and stability across cross-validation and test datasets.

- XGBoost achieved the best overall performance, effectively capturing non-linear relationships and complex feature interactions, which are often present in clinical data.
- Random Forest showed comparable results, highlighting its reliability and interpretability for healthcare applications.
- Logistic Regression, though slightly less accurate, remained a strong baseline model due to its simplicity, transparency, and robustness in clinical decision-making contexts.

The minimal difference between cross-validation and test results indicates strong generalization and absence of overfitting. These findings confirm that addressing data imbalance and applying ensemble learning significantly improve predictive performance in clinical severity classification tasks.

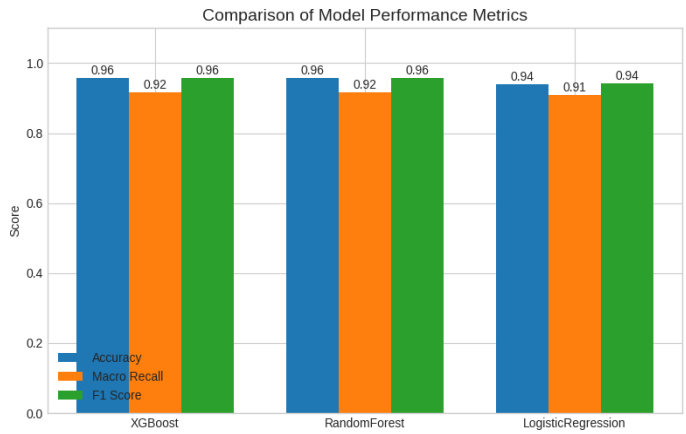

Figure 1Model comparison

Figure 2 illustrates the confusion matrix of XGBoost model. The model shows strong class-wise performance , accurately identifying the majority of Mild cases (102 of 105) and substantial proportion of moderate class(7 out of 9). Notably , the severe class is classified with complete accuracy.

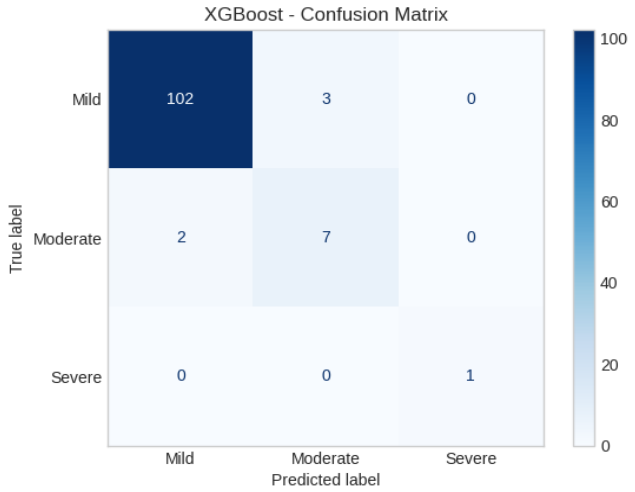

Figure 2:Confusion matrix (XGBoost)

In clinically decision support system, recall for severe class is prioritized over overall accuracy. Recall measures model's ability to identify all correctly predicted positive upon total actual positive. As Failure to detect severe cases may result in delayed or inadequate treatment. High Recall means fewer missed severe paitients[15]. The Recall(class-wise) for XGBoost model is 0.97, 0.78, 1.00 for mild moderate and severe classes respectively. The figure 2 confusion matrix confirms that the proposed model minimized such high-risk misclassification.

Class	Precision	Recall	F1-Score
Mild	0.98	0.97	0.98
Moderate	0.70	0.78	0.74
Severe	1.00	1.00	1.00

Precision, Recall, F1 for each class (XGBoost)

The class wise evaluation indicates strong performance for the Mild category , with high precision and recall , showing model's ability to correctly identify the majority class.
Importantly ,the Severe category gives perfect recall indication that high -risk cases were successfully identified without false negatives. The performance of Moderate class is comparatively lower , this can be attributed to overlapping lab profiles between adjacent severity level.

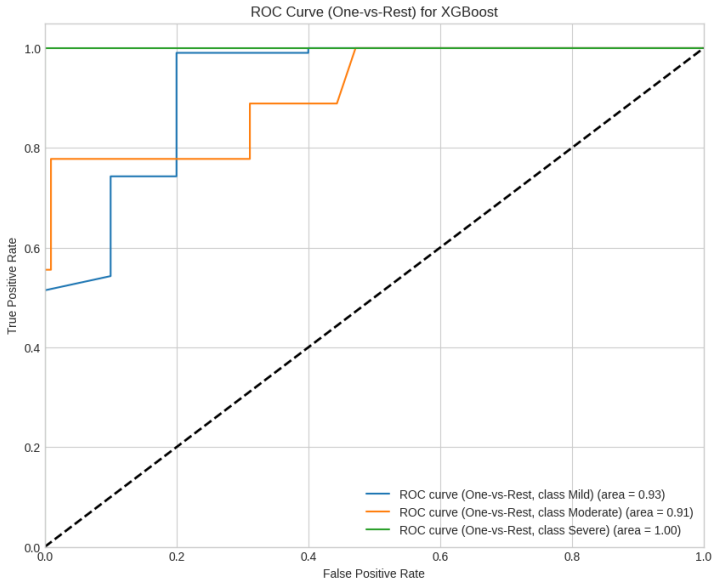

Figure 3One vs Rest ROC curves for XGBoost model across MIld , Moderate and Severe Severity classes

Fig.3. shows the One vs Rest ROC curves for the XGBoost classifier model. The model shows strong discriminative capability across all severity classes. TheAUC calues are 0.93 for Mild,0.91 for Moderate and 1.00 for Severe cases. This indicates very good separability between high risk patients and other severity categories.

6. Conclusion: -

This study developed a machine learning–based framework for predicting the severity of Sickle Cell Disease (SCD) using the MIMIC-IV clinical dataset. After addressing the issue of class imbalance through RandomOversampler, ensemble models such as XGBoost and Random Forest achieved superior and stable performance, with test accuracies and F1-scores around 95.7%–95.8%. These models effectively captured complex, non-linear relationships in patient data, while Logistic Regression maintained strong interpretability with slightly lower but consistent performance. Overall, the results demonstrate that integrating data balancing techniques with advanced ensemble learning can significantly enhance severity prediction accuracy and reliability in clinical decision support systems.

While the findings are encouraging, the study is constrained by the size and distribution of the available dataset. In particular, the relatively small number of Severe cases limits the strength of statistical conclusions for this category. As a result, further validation using larger and more diverse, multi-center datasets is necessary to confirm the stability and generalizability of the proposed framework. Despite these limitations, the results suggest that combining class balancing techniques with ensemble learning approaches offers a practical and promising direction for improving severity prediction and supporting data-driven clinical decision-making in SCD care.

Future Work:

Future efforts can focus on enhancing model robustness and clinical adoption through:

- Expanding the dataset across multiple hospitals to improve demographic diversity and generalization.
- Exploring deep learning architectures for better representation learning on complex clinical features.
- Integrating the model into clinical workflows through automated pipelines and decision-support dashboards.
- Validating model performance prospectively on real-world data to assess usability and safety in hospital environments.

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"The data was obtained from the MIMIC-IV database on PhysioNet after completing the required CITI training and adhering to a formal data use agreement. This ensured all ethical and privacy standards were met for the de-identified data."